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ORIGINAL ARTICLE



To suture or not to suture? Does globe immobilisation technique affect clinical outcome in stereotactic radiosurgery for uveal melanoma?

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ABSTRACT

Introduction: Stereotactic radiosurgery (SRS) is a valuable treatment option for uveal melanoma, offering excellent tumour control rates and eye preservation. Its efficacy relies upon accurate localisation of the tumour, which is challenging in the mobile eye. Various methods of globe immobilisation have been used, including non-invasive devices, such as eye movement tracking and suction cups, but common practice is to use local anaesthetic block with or without transconjunctival suturing of the extraocular muscles. Some studies have suggested that the addition of muscle suturing to local anaesthetic block provides better immobilisation of the globe, when compared to anaesthetic block alone. Controversy exists regarding the clinical relevance of this observation and ocular oncologists differ in their choice of immobilisation technique.

Methods: In order to establish if the addition of muscle suturing to local anaesthetic block improves clinical outcomes, we performed a retrospective review of all cases that underwent SRS for uveal melanoma over a 10-year period (May 2008 to May 2018). Based on surgeon preference, all patients received either local anaesthetic block plus muscle suturing (Group A) or local anaesthetic block alone (Group B) to induce globe akinesia. Outcomes assessed were primary treatment failure, tumour recurrence, secondary enucleation and death rate.

Results: In our cohort of 290 eyes; 118 patients were in group A and 172 patients were in group B. There were no cases of primary treatment failure in either group. With a minimum of 24 months follow-up, only 3 patients experienced tumour recurrence (1 in group A and 2 in group B). There was no significant difference in recurrence, enucleation and all-cause death rates between the two groups.

Conclusion: Our retrospective review suggests that although extraocular muscle suturing may be considered by some units to provide superior globe immobilisation for SRS, it does not alter the clinical outcome.

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Introduction

Stereotactic radiosurgery (SRS) is a valuable, proven treatment option for uveal melanoma, offering excellent rates of tumour control and eye preservation, if not vision preservation, in the majority of cases.^{1–7} SRS is especially useful for tumours which are unsuitable for alternative treatments due to the size and location of the tumour, and for patients who are not fit for general anaesthetic. The efficacy of the treatment is contingent upon accurate localisation of the radiation to the tumour, which is challenging in a mobile structure such as the eye.

The human eye is approximately 22–25 mm in diameter.⁸ It sits within the orbit and has an angular range of movement of approximately 88° horizontally, and 75° vertically.⁹ The eye can also move in an anterior-posterior direction due to gravity or raised pressure in the surrounding tissues, such as injection of fluid behind the eye. If such movements were to be unaccounted for, the tumour margin could be displaced outside the radiation field resulting in under-treatment, affecting both the accuracy and efficacy of SRS.

To prevent eye, and therefore tumour movement, various methods of globe immobilisation have been used. Non-invasive devices such as eye movement tracking¹⁰ and suction cups¹¹

allow detection of eye movements or physically prevent them. Retro- and peribulbar local anaesthetic blocks, either alone^{2,3} or in addition to transconjunctival suturing of the extraocular muscles^{4–7} are other methods, employed to temporarily paralyse the extraocular muscles and induce akinesia by anaesthetising the innervating cranial nerves.

At our centre, local anaesthetic block alone, or in combination with horizontal rectus muscle suturing are the only two fixation techniques utilised, and are the methods used most commonly elsewhere.¹² Local anaesthetic block alone has the benefit of being quicker and less invasive than the technique using sutures and does not require the additional step of suture removal post-SRS. Proponents of muscle suturing use this technique to mitigate posterior globe movement as the anaesthetic inevitably reabsorbs, a concern which is not directly addressed when using local anaesthetic alone.

Several studies have investigated eye position, using either repeated computerised tomography (CT) or magnetic resonance imaging (MRI), to investigate for movement of the eye between the planning stage of SRS and immediately post-treatment.^{2,4,13} One study using retrobulbar block alone indicated no significant shift of the globe,² whereas others have reported superior

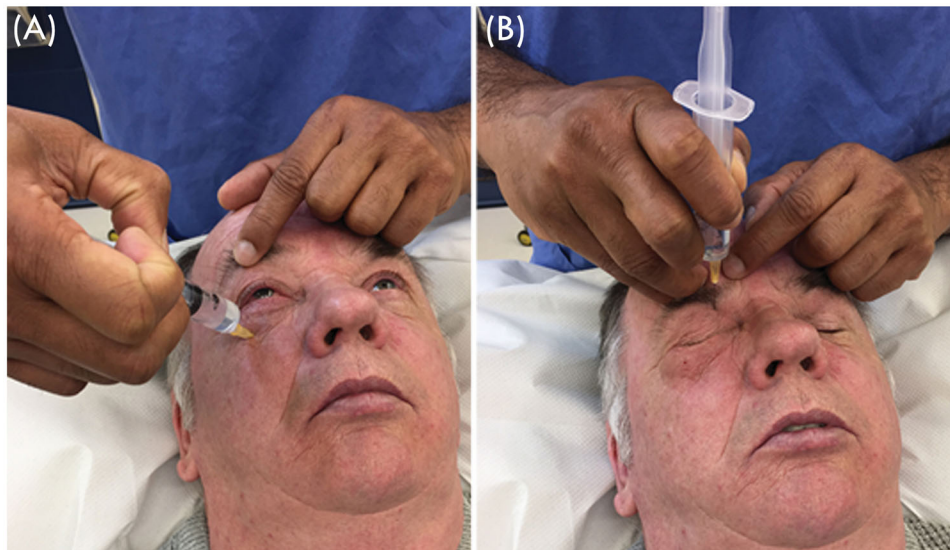


Figure 1. Patient undergoing periocular local anaesthetic injection (A) inferiotemporally and (B) superiolaterally to induce total globe akinesia without extraocular muscle suturing ahead of stereotactic radiosurgery for choroidal melanoma of the right eye.

globeimmobilisation^{4,13} and fewer complications¹² when extraocular muscle suturing was performed.

However, no published data has demonstrated different clinical outcomes in patients with uveal melanoma treated with SRS with different immobilisation techniques. In order to establish if the addition or omission of muscle suturing to local anaesthetic block alters outcomes in our practice, we performed a retrospective review of all cases treated with SRS for uveal melanoma at the National Centre for Stereotactic Radiosurgery, over a 10-year period.

Materials and methods

The Sheffield Ocular Oncology Service is a designated national adult ocular oncology service in the United Kingdom (UK) treating patients with uveal melanoma since 1985. Cases of intraocular melanoma are referred from different geographic locations, including England, Wales and Northern Ireland as well as from abroad. A robust database of all cases treated at our centre is continuously maintained by the database coordinator. Retrospective review was performed on all cases of uveal melanoma treated with Gamma Knife Stereotactic Radiosurgery over a 10-year period (May 2008 to May 2018).

Based on surgeon preference, patients received either local anaesthetic block plus muscle suturing (Group A) or periocular block alone (Group B) to produce total globe akinesia. Outcomes assessed were primary treatment failure, tumour recurrence, secondary enucleation and death rate. All cases of documented abandoned SRS due to eye movement were also recorded and examined for suture status. Statistical analysis was applied via Chi Square test for categorical data, and independent Student's *t*-test for numerical data.

In all cases, diagnosis of intraocular melanoma was made clinically via slit-lamp examination and indirect ophthalmoscopy with imaging techniques such as standardised A and B scan ultrasonography, ultrasound biomicroscopy and where appropriate, ocular coherence tomography scan and fundus angiography. The decision to perform stereotactic radiosurgery was based on tumour size and location, as well as patient co-morbidities and preference. All cases were discussed at the ocular oncology multidisciplinary team meeting prior to treatment.

Globe immobilisation and stereotactic surgical technique

Stereotactic radiosurgery was performed in all cases at the in-house National Centre for Stereotactic Radiosurgery, which was established over 35 years ago. All patients underwent globe immobilisation with local anaesthetic block, given either retrobulbar or peribulbar or as a combination (Figure 1). The local anaesthetic used was a combination of short acting 2% lignocaine and long acting 0.5% Bupivacaine. The use of hyaluronidase and/or 1:200,000 adrenaline was based on surgeon preference. A volume of between 8 and 15 ml was usually sufficient to produce complete akinesia of the eye. Where sutures were used, 3–0 silk on a 16 mm 3/8th cutting needle (Ethicon W502) was passed transconjunctivally through each horizontal rectus muscle in a single pass by grabbing the muscle through the translucent conjunctiva and tenons using non-toothed forceps and lifting the muscle slightly away from the scleral surface. The suture was then cut, and the other horizontal rectus muscle sutured similarly. The loose ends of each suture were tied firmly over a transparent plastic eye shield which was taped over the eye, effectively tenting the eye within the orbit.

The three ocular oncologists at our unit varied in their standard practice regarding the suturing of extraocular muscles: one (IGR) routinely sutured the horizontal rectus muscles, the second surgeon (PR) previously sutured all his SRS patients, and then changed his practice in 2013 and stopped suturing as he felt suturing probably added little benefit. The most recently appointed surgeon (SMS) has never used sutures.

Following globe immobilisation, all patients underwent fitting of a Leksell stereotactic frame under subcutaneous local anaesthetic, and MRI localisation of the tumour volume within an hour of globe immobilisation (where MRI was contraindicated, the patient underwent CT localisation). The treatment plan was correlated to the clinical and ocular imaging findings and the plan was checked and confirmed by the treating ocular oncologist and neurosurgeon. Once the planning stage was complete, patients would then be taken to the Leksell Gamma Knife, where they received a dose of 35 Gy to the 50% isodose delivered in a single session. The typical treatment volume for ocular melanomas (often fusiform in shape of around 350 mm³)

Table 1. Patient demographics and outcomes by suture status. * indicates statistically significant difference between the two groups ($p < 0.05$).

	Total	Gender		Mean age (years)	Mean follow up (months) [Range]	Tumour recurrence	Secondary enucleation	Death
		Male	Female					
Sutured (Group A)	118	78	40	65	92* [28 – 148]	1	7	23
Not Sutured (Group B)	172	103	69	68	53* [27 – 112]	2	14	34

is approximately 1700 mm³, and includes non-radiosensitive tissues such as vitreous humour and orbital fat, as well as a chorioretinal treatment margin of ~2 mm. Throughout, the eye is checked for movement by asking the patient to move their eyes – if movement was detected, treatment would be abandoned. All treatments were completed within 2.5 h of the administration of local anaesthetic, and within 60 minutes of the planning CT/MRI, as a day case procedure.

Results

292 patients underwent SRS for uveal melanoma during the study period with a minimum of 24 months follow-up; complete data was available for 290 (181 males, 109 females). The average age of patients at SRS was 67 years (range 19–94), and there was no significant difference in age between the two groups (Table 1). The mean follow-up for the cohort was 69 months (range 24–146).

Patients in group A (anaesthetic block plus sutures), had significantly longer follow-up (mean 92 months, range 26–146) compared to group B (block alone) (mean 53 months, range 24–110), $p < 0.00001$. This was due primarily to a change in practice by one surgeon (who performed the overwhelming majority of included cases) from routinely suturing all SRS patients to never suturing, meaning that many of those patients who were sutured were treated earlier in the study period. This difference is further compounded by the more recent appointment of a surgeon who does not suture his SRS patients.

Over the whole study group, there were no cases of primary treatment failure: all treated tumours showed signs of regression in the months following treatment. The recurrence rate in this study was 1.0% ($n = 3$). There was no significant difference in recurrence rate between group A (0.8%, $n = 1$) group B (1.2%, $n = 2$), $p = 0.8$. The secondary enucleation rate varied by suture status: 5.9% ($n = 7$) in group A, vs. 8.1% ($n = 14$) in group B, although this failed to reach statistical significance ($p = 0.5$). All secondary enucleations were performed for ischaemic complications following SRS, which are well-recognised complications of stereotactic radiosurgery for uveal melanoma.

Overall, the all-cause death rate was 19.7% with a mean of 67 months of follow-up and did not differ significantly by suture status. In those patients in group A, the rate of development of rubeosis (ocular neovascularisation due to radiation-induced ischaemia) was 18.6% ($n = 22$), in comparison to 14.0% ($n = 24$) in those in group B, again, not reaching statistical significance.

During the inclusion period, six patients had their treatment abandoned due to insufficient globe akinesia. These patients had undergone globe immobilisation and fitting of the stereotactic frame, however eye movement was detected before SRS was administered. Of these, 5 (83%) were immobilised with both local anaesthesia and suturing of the horizontal rectus muscles, and 1 patient (17%) underwent local anaesthetic block alone. This difference reaches statistical significance, $p = 0.036$, but is not indicative of causality. All patients subsequently had alternative treatment for their uveal melanoma.

Discussion/conclusion

Globe akinesia is crucial for treating uveal melanoma with SRS: it facilitates accurate localisation of radiation to the tumour and minimises collateral damage to adjacent structures. Debate exists as to the superiority of different immobilisation techniques, and specifically the necessity of extraocular muscle suturing in addition to local anaesthetic block.¹³ Several studies have investigated eye position both pre- and immediately post-SRS, with a view to answering this debate, with varying results.

Mueller's team² used retrobulbar anaesthesia alone and performed pre- and post-treatment MRI in 35 patients. They reported no 'significant shifting of the globe' when treatment was performed within 2.5 h of anaesthetic but did not report numerical data to quantify this. Tsai et al.⁴ assessed their technique of local anaesthesia plus suturing of the two horizontal extraocular muscles in a single patient with repeated CT scans, and found only 0.1 mm shift in globe position. Neither of these studies compared the two immobilisation techniques, but they suggest that either technique was successful at achieving akinesia.

Only one group compared both immobilisation techniques: Zubatkina et al.¹³ presented 87 patients who had either retrobulbar block alone, or in combination with rectus muscle suturing, and compared pre- and post-treatment globe position using MRI. The abstract from their presentation states that reliable immobilisation was only achieved with rigid suturing of the muscles, and that anaesthetic block alone resulted in 'significant shift'. However, as yet this study is unpublished and therefore further details are unavailable for quantification or further scrutiny.

An unpublished multicentre study¹² involving 349 patients from 8 centres across 7 countries suggested that local anaesthetic plus suturing was superior to local anaesthetic block alone. This included 84 cases that were sutured and 227 that were not, with a median of 40 months follow-up. The abstract states that extraocular muscle suturing was associated with significantly lower complication rates and vision loss, however the results have not been published via peer review.

Hence, to the best of our knowledge, our study is the first to compare clinical outcomes between the two most commonly employed globe immobilisation techniques used in SRS for uveal melanoma, involving such a large number of patients, from a single centre.

We present excellent local tumour control rates: 99.2% of those sutured (group A) and 98.8% of eyes immobilised with local anaesthetic alone (group B) achieved tumour control and remained recurrence-free for the duration of their follow-up. When interpreting these results, one must bear in mind the difference in the length of follow-up between the two groups, which was significantly longer in group A (92 months cf. 53 months, $p < 0.00001$). It is conceivable that further late recurrences could be identified with longer follow-up, unmasking a significant difference. However, we note that, all three recurrences in our study occurred within 42 months, which is well within the mean follow up period of group B.

Eye retention is also of importance when comparing immobilisation techniques, as secondary enucleation is a surrogate

measure of serious ocular complications from SRS treatment. There was no significant difference in the enucleation rate between the two groups in our study, (5.9% in group A, vs. 8.1% in group B, $p=0.5$), suggesting that immobilisation technique does not affect the risk of secondary enucleation. Although unlikely, the aforementioned difference in follow-up may have affected these results; it is possible that the secondary enucleation rate could increase with time. That being said, eye movement is not likely to cause increased ischaemic complications in the eye and therefore the authors do not believe suture-status should have an effect on secondary enucleation rates. Reassuringly, the enucleation rates in our study were not significantly different to those reported in our previous studies^{1,14} and others in the literature,^{2,5,7,12} suggesting that an increase in enucleation rate with time is not inevitable.

Locally, there had been concern by two co-authors that stopping muscle suturing could result in a larger number of abandoned procedures due to eye movement. In our experience, most abandoned treatments occurred due to unforeseeable treatment delays (of over 3 h) likely resulting in dissipation of the local anaesthetic effect, and was not related to extraocular muscle suturing, or the lack thereof.

Although this study is limited by its retrospective nature, our results suggest that local anaesthetic block alone achieves clinical outcomes which are as good as local anaesthetic plus muscle suturing. It may be that the dose we use (35 Gy to the 50% isodose) or the margins employed (~2 mm) mask a difference in efficacy caused by slight globe movement. However, our excellent local control rate (99%) and eye retention rate (94% with sutures, 92% without sutures) compare well with other published studies.^{1,2,5,7,12,14} Over the past 2 years we have reduced the dose delivered to 30 Gy to the 50% isodose to try and further reduce the risk of ischaemic complications and it will be important to see whether there is a commensurate rise in local recurrence rates in either the sutured or non-sutured group. This study is ongoing.

Ethical approval

This retrospective study was performed as a service evaluation and as such NHS Ethical Approval was not required.

Author contributions

HQ contributed to the design of the work, participated in data collection, analysis and interpretation, and wrote the manuscript. SMS participated in data analysis and interpretation and contributed to revision of the final manuscript. IGR revised the final manuscript. JY revised the final manuscript. MR contributed to the design of the work and revised the final manuscript.

PR contributed to the design of the work, participated in data collection, analysis and interpretation, and revised the final manuscript.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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